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Professional Summary

Backend engineer with **6 years** of experience building highly available, correctness-critical systems — payment engines, transactional ledgers, and data-intensive services in **Python**, **Go**, and **Java**. Deep hands-on with **Python + FastAPI + MongoDB + Redis + Airflow** from shipping a startup platform to **100% uptime** as team lead and scaling a matching service to **20,000+ DAU**; currently engineer a **5,000+ QPS** payment engine at Toshiba TEC. Fluent in English (business), Japanese (business), and native Chinese — comfortable with day-to-day written English collaboration with product and cross-functional teams.

Professional Experience

Backend Engineer — Next-Gen Enterprise POS & Payment Engine

Aug 2025 – Present

Toshiba TEC · Tokyo, Japan

- Architecting a high-availability core payment engine benchmarked to sustain **5,000+ peak QPS**, routing high-frequency retail transactions with the same correctness and traceability guarantees required by billing and claims-processing pipelines.
- Guaranteed transaction idempotency and eliminated double-charge race conditions during concurrent C2B checkouts by implementing row-level pessimistic locking (`SELECT ... FOR UPDATE`) — a pattern directly transferable to duplicate-claim prevention in RCM workflows.
- Designed an event-driven architecture on the **Transactional Outbox Pattern** + **Kafka** to decouple the core ledger from downstream services, ensuring theoretical zero data loss across asynchronous status updates.
- Defined strict service contracts via **gRPC/Protobuf** and integrated debounce + state-polling on the client to keep the pipeline responsive under network latency.

Senior Backend Engineer / Team Lead — Job-Matching Platform & Ticketing

Apr 2023 – Jul 2025

System

ServiceMeme Inc. · Tokyo, Japan

- Led a cross-functional team of **4** to deliver a scalable job-matching platform on **Python + FastAPI** with **MongoDB** as the primary store and **Redis** for caching and task queues — sustained **100% uptime** through major feature rollouts.
- Accelerated search query performance by **80%** by redesigning **MongoDB** collections and compound indexes around real access patterns, resolving long-standing legacy bottlenecks.
- Orchestrated batch enrichment and matching pipelines with **Apache Airflow** DAGs — nightly candidate scoring, index rebuilds, and analytics rollups running under strict SLAs.
- Owned end-to-end backend design in a startup environment: API contracts with PMs (written English), data modeling, containerized deploys (Docker + AWS), on-call, and post-mortems.
- Architected an amusement-park ticketing platform as **Python + FastAPI** microservices to handle high-concurrency online sales, using **Redis** for distributed locks and rate limiting, backed by **MongoDB**.
- Shipped a unified department-store payment gateway on aggressive timelines, resolving complex legacy integrations across multiple business units via a **Python + FastAPI** service backed by **MongoDB**.

Backend Engineer — Community & E-Commerce Platform

Oct 2021 – Mar 2023

QianDao App · Beijing, China

- Scaled a LARP matchmaking service DAU from **4k to 20k (+400%)** by engineering high-performance matching algorithms in **Go (Gin)** + **MySQL**, with **Redis** for hot-path caching and rate control.
- Architected a scalable UGC tag and rating system over **Standard Product Unit (SPU)** data in a distributed microservice environment, supporting hyper-growth and rapid product iteration.

Backend Engineer — FusionSolar Software

Jul 2020 – Sep 2021

HUAWEI · China

- Developed core control-system modules (fan-control, low-power mode) in **Python**, enabling the on-time global market launch of a flagship enterprise solar-inverter product.
- Coordinated with downstream engineering and hardware teams to deliver robust remote monitoring capabilities and streamline mass-production testing cycles.

Technical Stack

Languages	Python · Go · Java · TypeScript / JavaScript · PHP
Backend & Frameworks	FastAPI · Django · Spring Boot · Gin · Node.js · Laravel
Databases & Storage	MongoDB · Redis · PostgreSQL · MySQL · DynamoDB · Elasticsearch / OpenSearch
Data & Orchestration	Apache Airflow · Kafka · Event-driven pipelines · Outbox Pattern · Serverless (AWS Lambda)
Architecture	Microservices · REST · gRPC / Protobuf · Idempotency & concurrency control · Distributed locking
Cloud & DevOps	AWS (ECS, Lambda, RDS) · GCP · Docker · GitHub Actions · CI/CD
Spoken Languages	English (Business) · Japanese (Business) · Chinese (Native)

Education

B.Sc. in Computer Science and Technology

Sep 2016 – Jun 2020

Beijing Forestry University